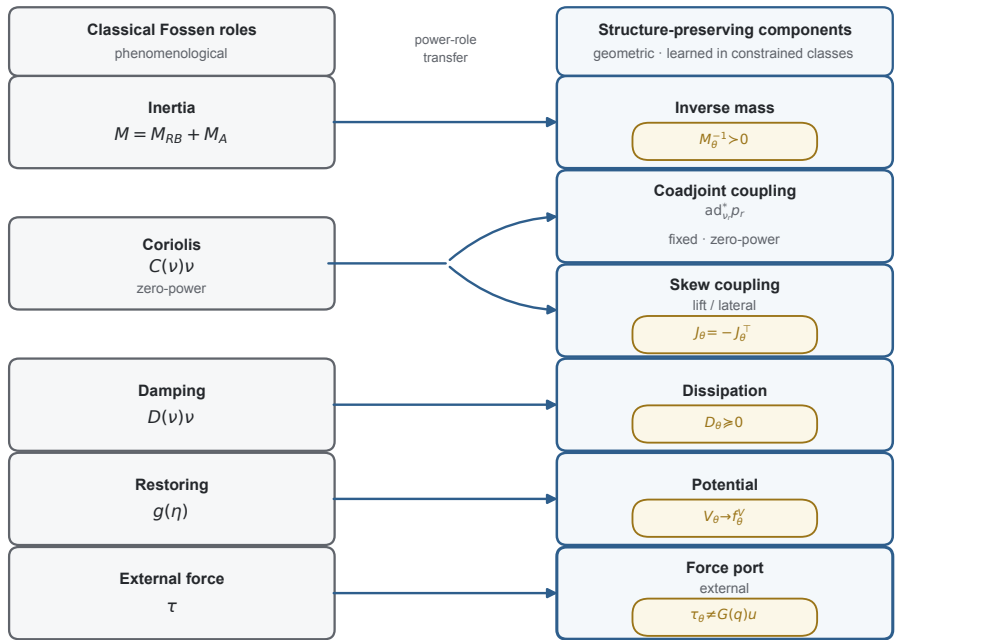




explicit force sum

$$M\dot{v} + C(v)v + D(v)v + g(\eta) = \tau$$



open structured form

$$\dot{p}_r = \text{ad}_{v_r}^* p_r + f_\theta^V - D_\theta(\xi)v_r + J_\theta(\xi)v_r + \tau_\theta$$

power-pairing velocity $v \rightarrow v_r$ (relative-water)

external port

blue = structural prior · gold = data-learned shape

Mechanical-core power roles; current / actuator / depth are exogenous. The full vehicle–actuator–environment system is open — not a closed port-Hamiltonian system, and $\tau_\theta \neq G(q)u$.